



Smart Asset Clustering

SMARTGALLERY DAM FOR COMFYUI — COMPLETE ILLUSTRATED MANUAL

Welcome to the official user manual for **Smart Asset Clustering**. This guide will help you understand what asset clustering is, how to use it step-by-step, and how to leverage it to organize, compare, and audit your ComfyUI generation library.

1. What is Smart Asset Clustering?

When working in ComfyUI, creating artwork is an iterative process. You often generate dozens or hundreds of images and videos while tweaking seeds, refining prompts, adjusting LoRA weights, or testing different checkpoints. Over time, your gallery can become cluttered with similar-looking renders spread across multiple folders.

Smart Asset Clustering automatically organizes your gallery by analyzing the generation recipe embedded in your media files. Without requiring any manual tagging, it groups your renders into logical clusters based on either their **node architecture** or their **prompt text**.

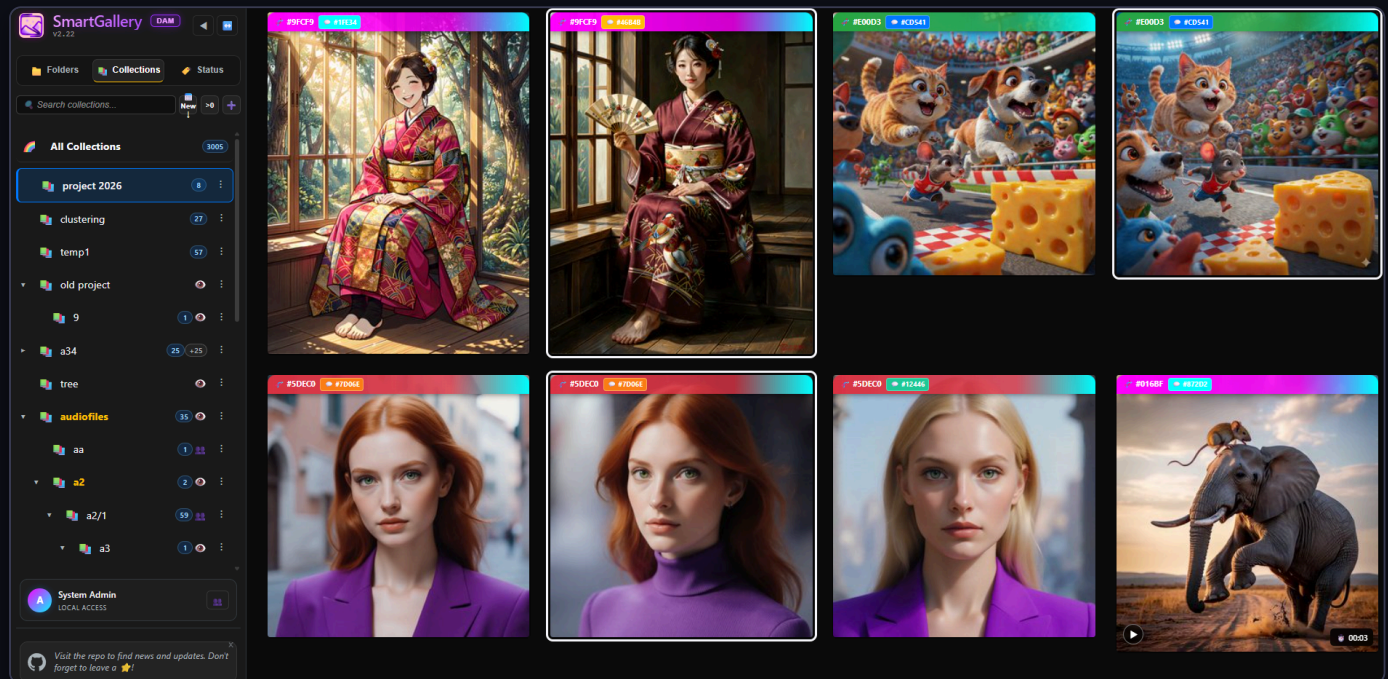


Figure 1: SmartGallery main gallery grid displaying color-coded cluster hash badges on media thumbnails.

2. The Two Clustering Modes

Smart Asset Clustering offers two primary ways to group your media:

⚙️ 1. Architecture Clustering (Workflow Graph)

- **What it does:** Groups all images and videos created using the exact same node connections and pipeline structure.
- **What it ignores:** Ignores random seeds, prompts, step counts, CFG scale, and noise settings.
- **When to use it:** When you want to see all outputs generated by a specific ComfyUI workflow template.

💬 2. Prompt Text Clustering

- **What it does:** Groups all images and videos sharing the exact same positive prompt text.
- **What it ignores:** Ignores node graphs, samplers, checkpoints, and LoRAs.
- **When to use it:** When you want to compare how different models, checkpoints, or LoRAs render the exact same text prompt.

3. Search Scope & Inner Sorting

When configuring a cluster search, you can customize the scope and order in the modal dialog:

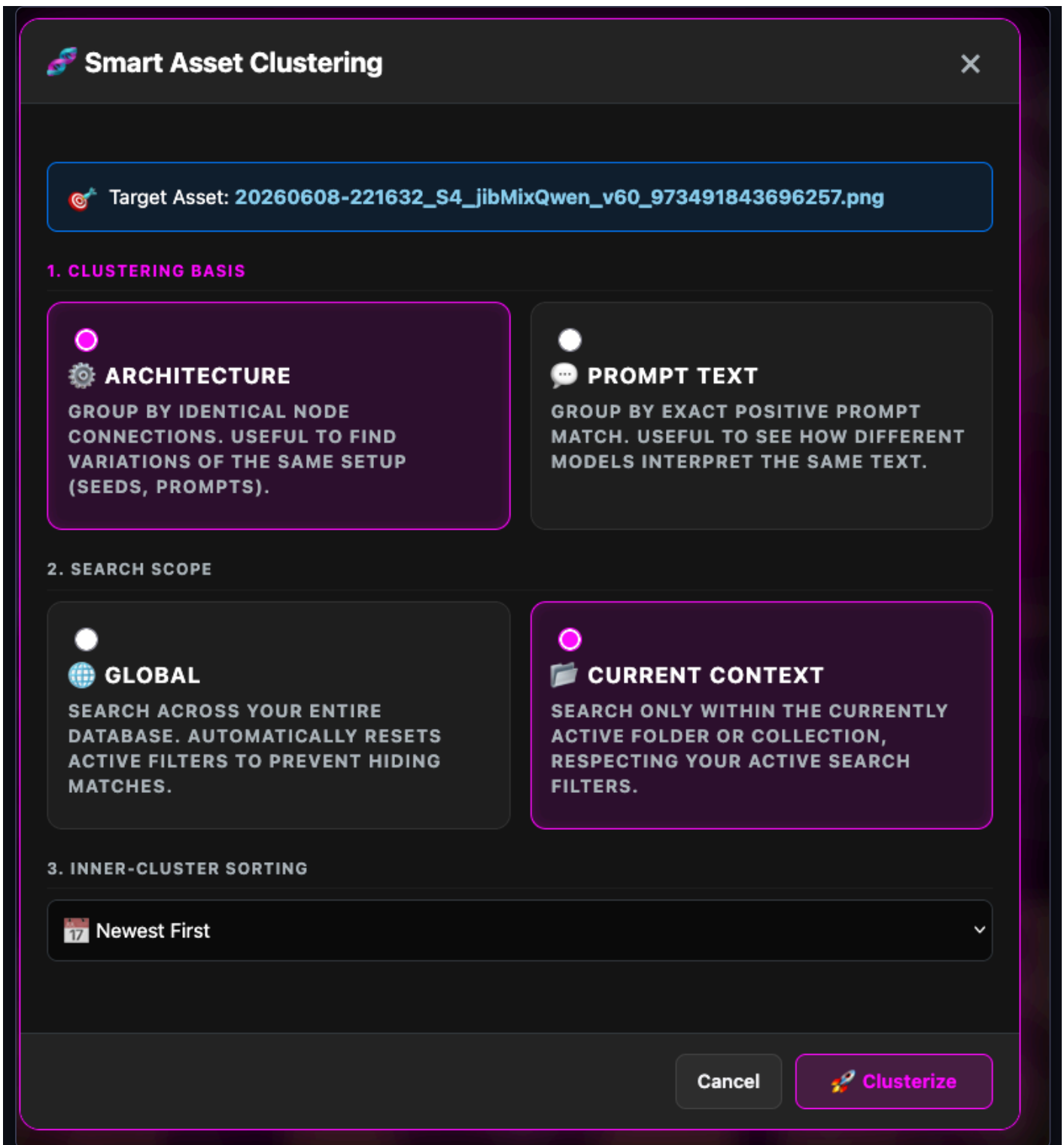


Figure 2: Smart Asset Clustering configuration modal showing basis selection, search scope, and sorting options.

Search Scope

-  **Global Scope:** Searches your entire library across all folders and drives. It automatically clears active search filters to ensure every matching asset is discovered.
-  **Current Context:** Restricts the search to your currently active folder or virtual collection, allowing targeted clustering within a specific project.

Inner-Cluster Sorting

Controls how assets are ordered within each cluster group:

-  **Newest First:** Shows the most recently generated renders at the top of each cluster.
-  **Oldest First:** Shows the earliest renders at the top of each cluster.

- **★ Highest Rated First:** Orders items within each cluster by their DAM star rating score.

4. How to Use Smart Asset Clustering (Step-by-Step)

Method 1: Using the Keyboard Shortcut (**Shift + C**)

1. In the main gallery grid or Lightbox, hover your mouse over any image or video (or highlight it with keyboard arrows).
2. Press **Shift + C** on your keyboard.
3. The **Smart Asset Clustering** window will open, automatically targeting the selected file.
4. Choose your **Clustering Basis** (Architecture or Prompt Text) and **Scope**.
5. Click **Clusterize**.

Method 2: From the Tools Menu

1. Click **Tools** in the top navigation bar.
2. Select **Smart Clustering**.
3. Configure your desired options in the dialog window.
4. Click **Clusterize**.

5. Navigating Active Cluster Mode

Once clustering is activated:

- **Active Cluster Banner:** A bright fuchsia banner appears at the top of your gallery indicating that Clustering Mode is active, showing the current basis, scope, and total matching assets.
- **Cluster Badges:** Each thumbnail displays a distinct color-coded hash badge (e.g., **#5DEC0** or **#7D06E**). Items sharing identical color badges belong to the exact same cluster.
- **Exiting Cluster Mode:** Click the red **Exit Cluster Mode** button on the top banner, or press **ESC** on your keyboard at any time.

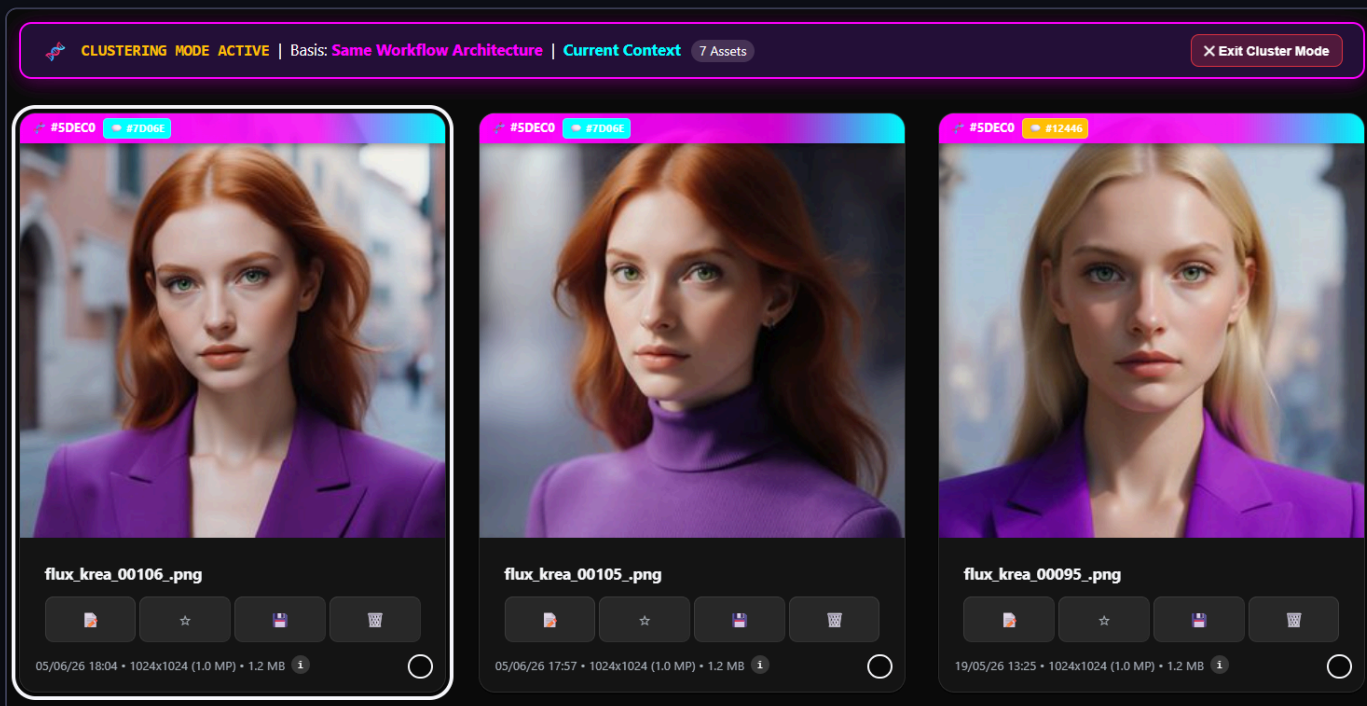


Figure 3: Active Clustering Mode grid view filtering gallery items by identical ComfyUI workflow architecture.

6. Deep Inspection with the Cluster Inspector & Info Panel

SmartGallery DAM provides two distinct ways to inspect cluster metadata: through dedicated Modal Inspectors and via the Asset Info Panel tab.

A. Modal Inspectors

Clicking on any cluster badge on a thumbnail opens the **Cluster Inspector Modal**, offering a comprehensive overview of the node sequence, models loaded, and prompt text.

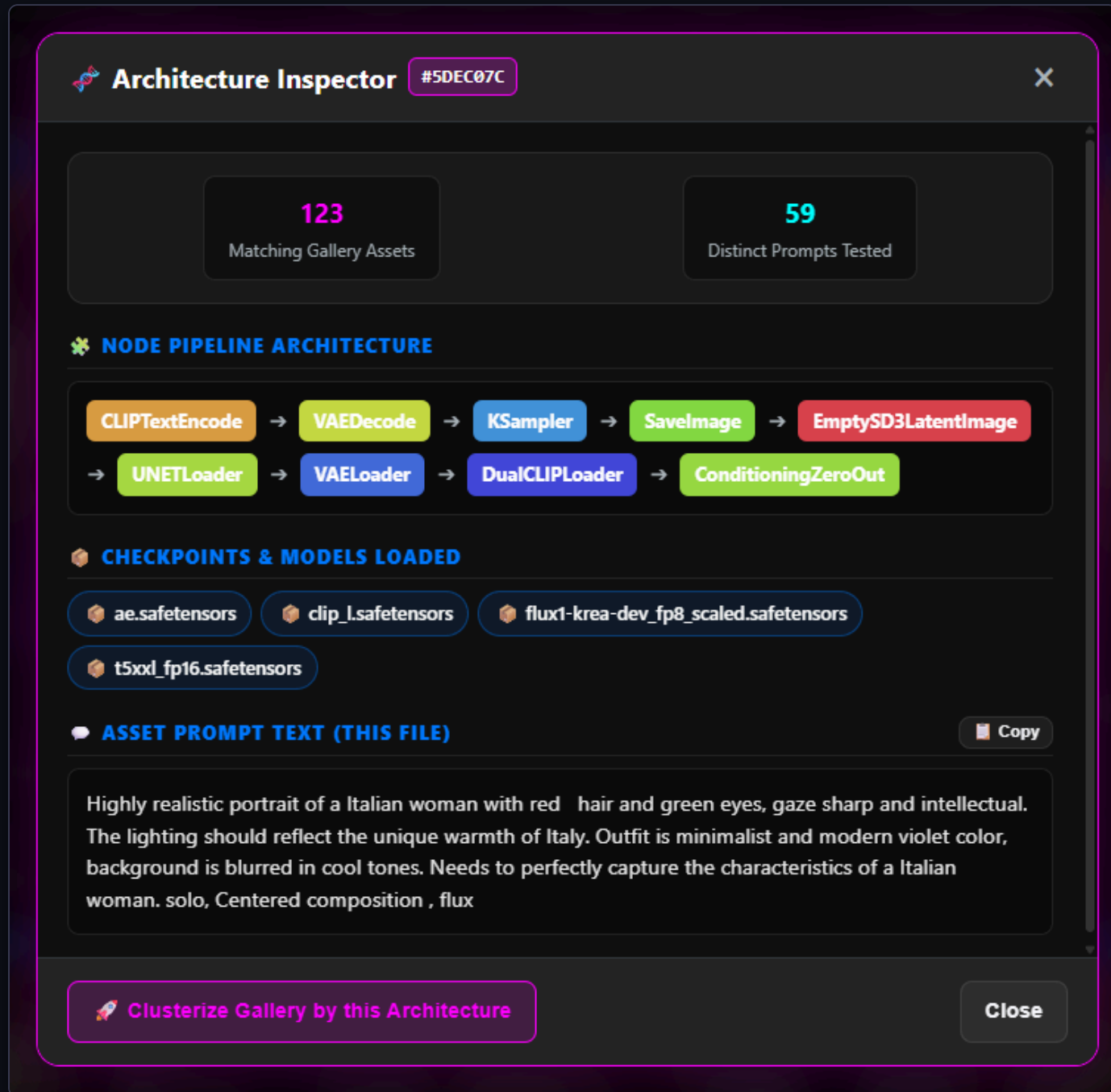


Figure 4: Architecture Inspector Modal displaying total matching assets, distinct prompt variations, node flow sequence, and loaded checkpoints.

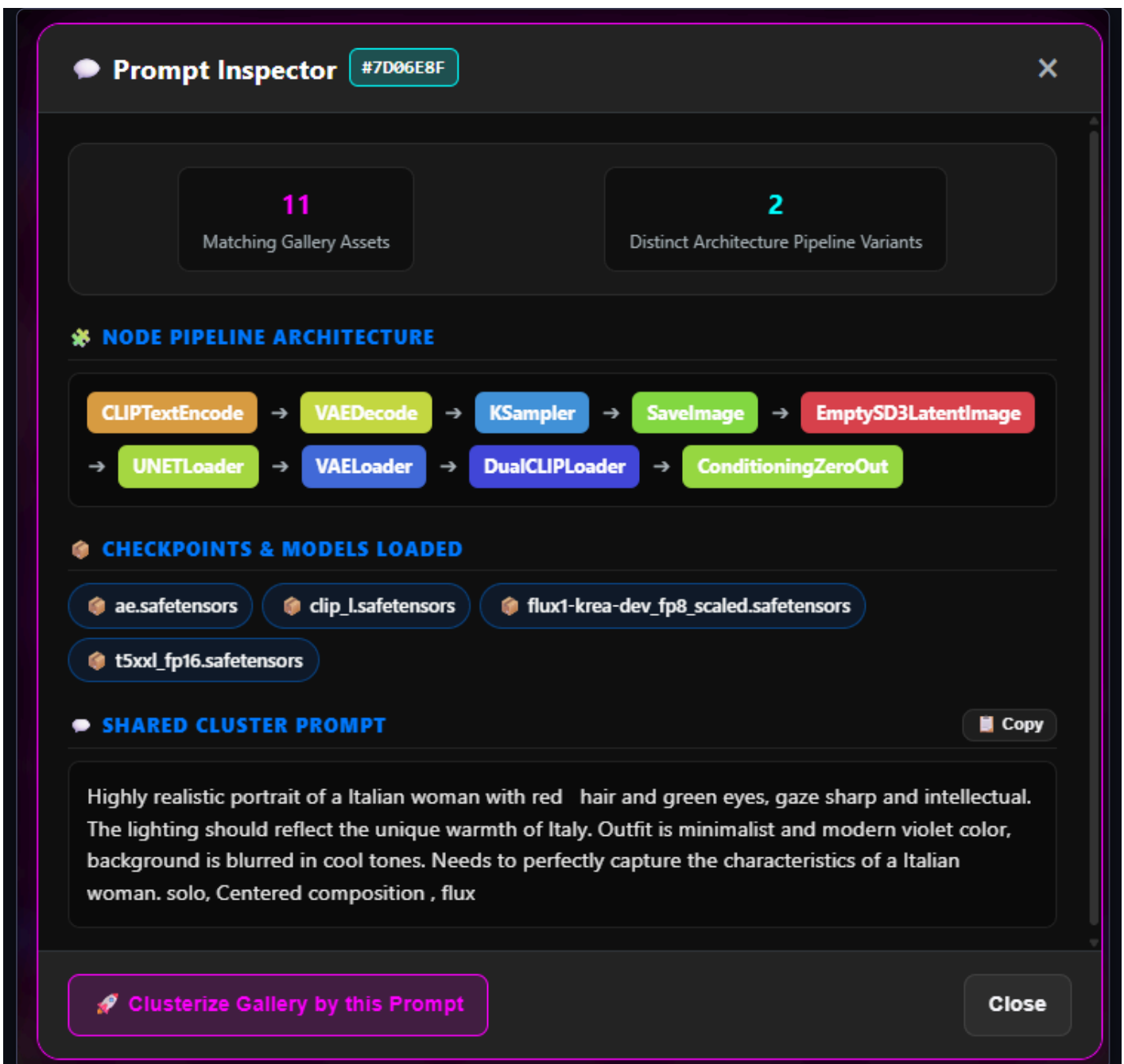


Figure 5: Prompt Inspector Modal showing matching assets, distinct architecture variants, node pipeline, and shared prompt text.

B. Side Info Panel (Architecture & Cluster Tab)

When inspecting any individual media file, navigating to the **Architecture & Cluster** tab in the side panel provides quick one-click cluster hashes and an itemized breakdown of the asset's generation workflow.



7. Practical Use Cases & Scenarios

Scenario 1: Model & LoRA Comparison Testing

Goal: You want to test how 5 different checkpoints render the prompt "A futuristic cyberpunk street in neon rain".

1. Generate renders across your checkpoints using that exact prompt.
2. Select one of the images and press **Shift + C**.
3. Choose  **Prompt Text** as the basis and set scope to  **Global**.
4. Click  **Clusterize**.

Result: SmartGallery gathers all 5 renders together into a single cluster, letting you compare the artistic output side-by-side.

Scenario 2: Finding Seed Variations of a Favorite Render

Goal: You built an upscale pipeline three weeks ago and generated 30 seed variations scattered across folders.

1. Open your favorite render in the Lightbox and press **Shift + C**.
2. Select  **Architecture** as the basis and set scope to  **Global**.
3. Click  **Clusterize**.

Result: SmartGallery instantly locates all 30 renders across your entire hard drive that used that exact node pipeline.

Scenario 3: Folder Auditing & Cleaning

Goal: You want to clean up a project folder and organize your renders by workflow type before sending choices to a client.

1. Navigate to your project folder.
2. Open Tools →  **Smart Clustering**.
3. Select  **Architecture** and set scope to  **Current Context**.
4. Set Inner-Cluster Sorting to  **Highest Rated First**.

Result: Renders in your active folder are grouped by workflow structure, with top-rated images displayed first.

8. Keyboard Shortcuts Summary

Shortcut	Context	Action
Shift + C	Gallery Grid / Lightbox	Open Clustering Modal for the active or hovered asset
ESC	Active Clustering Mode	Instantly exit Clustering Mode and return to normal gallery view
ESC	Modals / Inspector	Close the Cluster Inspector window
Click Badge	Grid Card	Open the Cluster Inspector for that specific cluster hash

SmartGallery DAM — Organize your creative pipeline effortlessly.